

Отчёт по лабораторной работе 7

Адресация IPv4 и IPv6. Настройка DHCP

Талебу Тенке Франк Устон

Российский университет дружбы народов, Москва, Россия

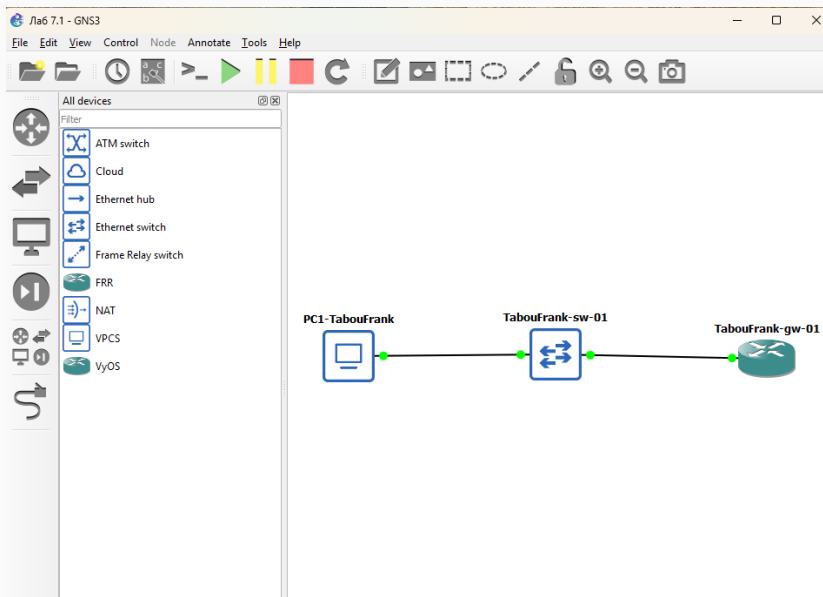
Цели и задачи

Цель лабораторной работы

Получение навыков настройки службы DHCP на сетевом оборудовании для распределения адресов IPv4 и IPv6.

Выполнение лабораторной работы

Топология стенда (IPv4)



Базовая настройка VyOS

Лаб 7.1 - GNS3

File Edit View Control Node Annotate Tools Help

Запустить и обновить

Таблица 7.1

DNS
10.0.0.1 (DHCP)

0.0.0.1, DHCP-10.0.0.253, на-зовать пакеты

соответствии-тизатор VyOS

присвойте на-

лизатор Wireshark [Taboufrank-sw-01 Ethernet1 to Taboufrank-gw-01 eth0]

Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Фильтр отображения ... <Ctrl-/>

All devices

Filter

- ATM switch
- Cloud
- Ethernet hub
- Ethernet switch
- Frame Relay switch
- FRR
- NAT
- VPCS
- VyOS

PC1-Taboufrank

Taboufrank-sw-01

Taboufrank-gw-01

```
graph LR; PC1[PC1-Taboufrank] --- S1(( )); S1 --- S2(( )); S2 --- S3(( )); S3 --- GW[Taboufrank-gw-01];
```

Настройка DHCP-сервера (IPv4)

```
h>Welcome to VyOS!
r
Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev
R
You can change this banner using "set system login banner post-login" command.
IA
VyOS is a free software distribution that includes multiple components,
p you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
y You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set ssystem host-name TabouFrank-gw-01

Configuration path: [ssytem] is not valid
Set failed

[edit]
vyos@vyos# set system host-name TabouFrank-gw-01
[edit]
vyos@vyos# set system domain-name TabouFrank.net
[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password
TabouFrank

Configuration path: system login user TabouFrank [authentication] is not val
id
Set failed

[edit]
vyos@vyos# set system login user TabouFrank authentication plaintext-password d
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$
```

Получение адреса на клиенте (PC1)

```
Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 4: Получение IP-адреса по DHCP на PC1

Проверка связности (IPv4)

```
Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01#
```

Рис. 5: Проверка IPv4-конфигурации и ping

Статистика и аренды DHCP (IPv4)

```
Welcome to VyOS - TabouFrank-gw-01 ttyS0

TabouFrank-gw-01 login: TabouFrank
Password:
Welcome to VyOS!

Check out project news at https://blog.vyos.io
and feel free to report bugs at https://vyos.dev

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
TabouFrank@TabouFrank-gw-01:~$ configure
[edit]
TabouFrank@TabouFrank-gw-01# delete system login user vyos
[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24
[edit]
nk domain-name TabouFrank.net

Configuration path: service dhcp-server [shared-network-nae] is not valid
Set failed

[edit]
nk domain-name TabouFrank.net et service dhcp-server shared-network-nae TabouFrank
nk domain-name TabouFrank.net et service dhcp-server shared-network-name TabouFrank
[edit]
nk name-server 10.0.0.1-01# set service dhcp-server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 default-router 10.0.0.1 server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 range hosts start 10.0.0.2 server shared-network-name TabouFrank
[edit]
nk subnet 10.0.0.0/24 range hosts stop 10.0.0.253
[edit]
TabouFrank@TabouFrank-gw-01#
```

Журнал DHCP и этапы обмена

```
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# save
Warning: you have uncommitted changes that will not be saved.

Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# commit

Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
[ service dhcp-server ]
None of the configured subnets have an appropriate primary IP address on
any broadcast interface configured, nor was there an explicit listen-
address configured for serving DHCP relay packets!

[[service dhcp-server]] failed
Commit failed
[edit]
TabouFrank@TabouFrank-gw-01# address dhcp

Invalid command: [address]

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands |match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# delete interfaces ethernet eth0 address dhcp
[edit]
TabouFrank@TabouFrank-gw-01# set interfaces ethernet eth0 address 10.0.0.1/24

Configuration path: [interfaces ethernet eth0 address 10.0.0.1/24] already exists

[edit]
TabouFrank@TabouFrank-gw-01# show configuration commands | match eth0

Configuration path: [configuration] is not valid
Show failed

[edit]
TabouFrank@TabouFrank-gw-01# commit
[edit]
TabouFrank@TabouFrank-gw-01#
```

Анализ захваченного трафика (DHCP)

```
TabouFrank@TabouFrank-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
TabouFrank@TabouFrank-gw-01# exit
exit
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases   Available  Usage
-----
TabouFrank 252      0        252      0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start  Lease expiration  Remaining  Pool  Hostname
-----
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 8: Анализа DHCP

Настройка IPv6

Расширение топологии (IPv6)

```
TabouFrank@TabouFrank-gw-01:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           10.0.0.1/24      u/u
eth1           -                u/D
eth2           -                u/D
lo             127.0.0.1/8      u/u
              ::1/128
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 9: Расширенная топология лабораторного стенда

Настройка IPv6-адресов на VyOS

```
VPCS> ip dhcp -d
Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 1 (REQUEST)
Client IP Address: 0.0.0.0
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Discover
Option 12: Host Name = VPCS
Option 61: Client Identifier = Hardware Type=Ethernet MAC Address = 00:50:79:66:
68:00

Opcode: 2 (REPLY)
Client IP Address: 0.0.0.0
Your IP Address: 10.0.0.2
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Offer
Option 54: DHCP Server = 10.0.0.1
Option 51: Lease Time = 86400
Option 1: Subnet Mask = 255.255.255.0
Option 3: Router = 10.0.0.1
Option 6: DNS Server = 10.0.0.1
Option 15: Domain = TabouFrank.net

Opcode: 1 (REQUEST)
Client IP Address: 10.0.0.2
Your IP Address: 0.0.0.0
Server IP Address: 0.0.0.0
Gateway IP Address: 0.0.0.0
Client MAC Address: 00:50:79:66:68:00
Option 53: Message Type = Request
Option 54: DHCP Server = 10.0.0.1
```

Настройка RA + DHCPv6 Stateless (eth1)

```
VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 10.0.0.2/24
GATEWAY     : 10.0.0.1
DNS         : 10.0.0.1
DHCP SERVER : 10.0.0.1
DHCP LEASE  : 86264, 86400/43200/75600
DOMAIN NAME : TabouFrank.net
MAC         : 00:50:79:66:68:00
LPORT      : 10054
RHOST:PORT  : 127.0.0.1:10055
MTU         : 1500

VPCS> ping 10.0.0.1 -c 2

64 bytes from 10.0.0.1 icmp_seq=1 ttl=64 time=3.137 ms
64 bytes from 10.0.0.1 icmp_seq=2 ttl=64 time=2.496 ms

VPCS> 
```

Рис. 11: Настройка Router Advertisements и DHCPv6 Stateless

Проверка конфигурации сервисов на VyOS

```
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available  Usage
-----
TabouFrank 252      1         251        0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server statistics
Pool      Size    Leases    Available  Usage
-----
TabouFrank 252      1         251        0%
TabouFrank@TabouFrank-gw-01:~$ show dhcp server leases
IP address  Hardware address  State  Lease start      Lease expiration  Remaining  Pool      Hostname
-----
10.0.0.2    00:50:79:66:68:00 active  2026/01/03 23:50:18  2026/01/04 23:50:18  23:51:55  TabouFrank  VPCS
TabouFrank@TabouFrank-gw-01:~$
```

Рис. 12: Просмотр конфигурации DHCPv4 и DHCPv6

Проверка клиента PC2 (Stateless)

Захват из Standard input [TabouFrank-sw-01 Ethernet1 to TabouFrank-gw-01 eth0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
246	4178.347772	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x875e35
247	4183.787185	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x875e35
248	4194.069897	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x875e35
249	4214.976866	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x875e35
250	4233.039940	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x875e35
251	4296.885829	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
252	4302.592174	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
253	4311.248619	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
254	4320.414823	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
255	4335.146832	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
256	4353.380318	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xe5701a2f
257	4434.640433	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
258	4439.104756	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
259	4443.488667	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
260	4447.740925	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
261	4458.571266	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
262	4474.900938	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
263	4493.115709	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x461dae55
264	4822.664598	0.0.0.0	255.255.255.255	DHCP	406	DHCP Discover - Transaction ID 0xbd3c0452
265	4822.670141	0c:31:84:6b:00:00	Broadcast	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1

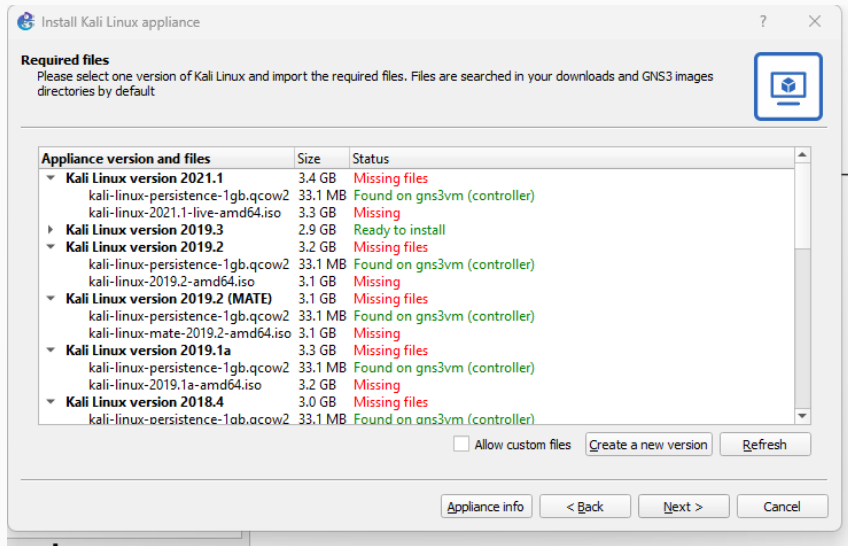
Checksum: 0x9234 [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 247]
[Timestamps]
UDP payload (300 bytes)
Dynamic Host Configuration Protocol (Discover)
Message type: Boot Request (1)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x461dae55
Seconds elapsed: 23
Bootp flags: 0x0000 (Unicast)
Client IP address: 0.0.0.0
Your (client) IP address: 0.0.0.0
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: 0c:31:84:6b:00:00 (0c:31:84:6b:00:00)
Client hardware address padding: 000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (Discover)
Option: (12) Host Name

0000 ff ff ff ff ff ff 0c 31 84 6b 00 00 08 05 101
0010 01 48 00 00 00 00 00 11 39 96 00 00 00 ff ffH.....5
0020 ff ff 00 44 00 43 01 34 92 34 01 01 06 00 46 1dD.C.4
0030 ae 55 00 17 00 00 00 00 00 00 00 00 00 00 00U.....
0040 00 00 00 00 00 00 0c 31 84 6b 00 00 00 00 001
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0090 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00a0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00b0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00c0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00d0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00e0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00f0 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0100 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0110 00 00 00 00 00 00 63 82 53 63 35 01 01 0c 10 54c.S
0120 61 62 6f 75 46 72 61 6e 6b 2d 67 77 2d 30 31 37aboutFran
0130 07 01 1c 03 06 79 0f 1a ff 00 00 00 00 00 00y.....
0140 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0150 00 00 00 00 00 00 00 00 00 00 00 00 00 00
aboutFran

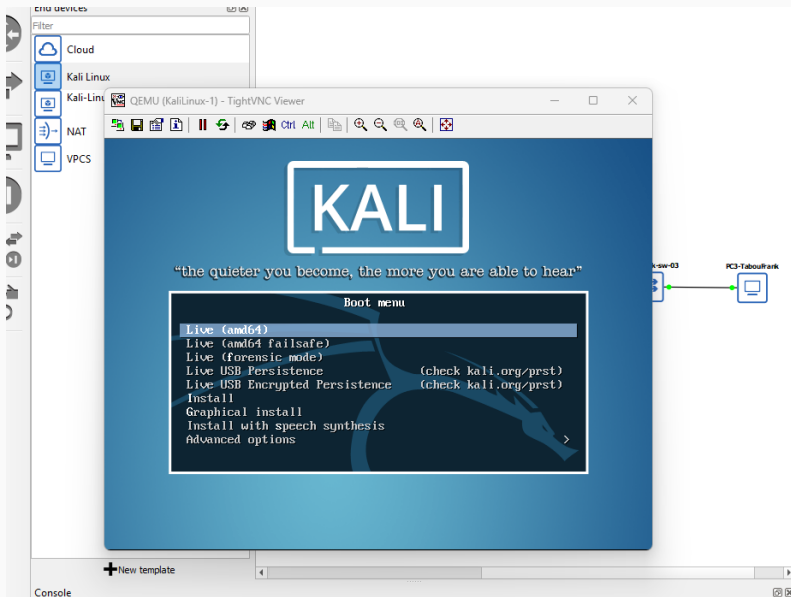
Проверка связности (IPv6)

```
TabouFrank@TabouFrank-gw-01:~$ show log | grep dhcp
Jan 03 22:41:51 dhclient-script-vyos[1846]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:41:52 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:41:52 dhclient-script-vyos[1846]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:41:52 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[2089]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:43:47 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:43:47 dhclient-script-vyos[2089]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:43:47 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:45:55 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:45:55 dhclient-script-vyos[2139]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:45:55 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2173]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:47:42 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:47:42 dhclient-script-vyos[2173]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:47:42 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:49:20 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:49:20 dhclient-script-vyos[2221]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:49:20 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:50:59 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:50:59 dhclient-script-vyos[2271]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:50:59 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:53:23 dhclient-script-vyos[2320]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:53:23 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:53:23 dhclient-script-vyos[2320]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:53:23 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:55:47 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:55:47 dhclient-script-vyos[2371]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:55:47 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:58:05 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 22:58:05 dhclient-script-vyos[2427]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 22:58:05 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:00:20 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:20 dhclient-script-vyos[2489]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:00:20 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:00:57 sudo[2607]: TabouFrank : TTY=ttys0 ; PWD=/home/TabouFrank ; USER=root ; COMMAND=/usr/bin/sh -c /usr
z/sbin/vysshim /usr/libexec/vyos/conf_mode/dhcp_server.py
Jan 03 23:00:57 vyos-configd[610]: Received message: {"type": "node", "data": "/usr/libexec/vyos/conf_mode/dhcp_se
rver.py"}
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:02:07 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:02:07 dhclient-script-vyos[2646]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:02:07 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2680]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:03:53 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:03:53 dhclient-script-vyos[2680]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:03:53 vyos-hostsd[608]: Request data: {"type": "name_servers", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:06:23 dhclient-script-vyos[2714]: Deleting search domains with tag "dhcp-eth0" via vyos-hostsd-client
Jan 03 23:06:23 vyos-hostsd[608]: Request data: {"type": "search_domains", "op": "delete", "data": [{"dhcp-eth0"}]}
Jan 03 23:06:23 dhclient-script-vyos[2714]: Deleting nameservers with tag "dhcp-eth0" via vyos-hostsd-client
```

Получение параметров по DHCPv6 (Stateless)



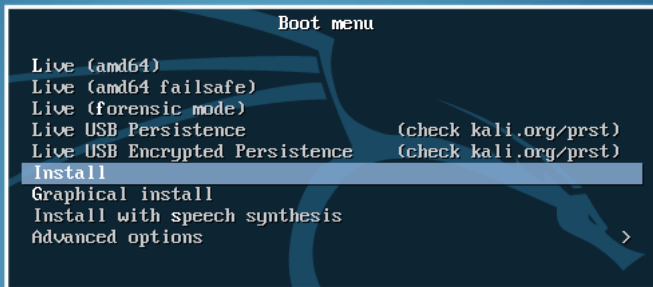
Аренды DHCPv6 в Stateless-режиме



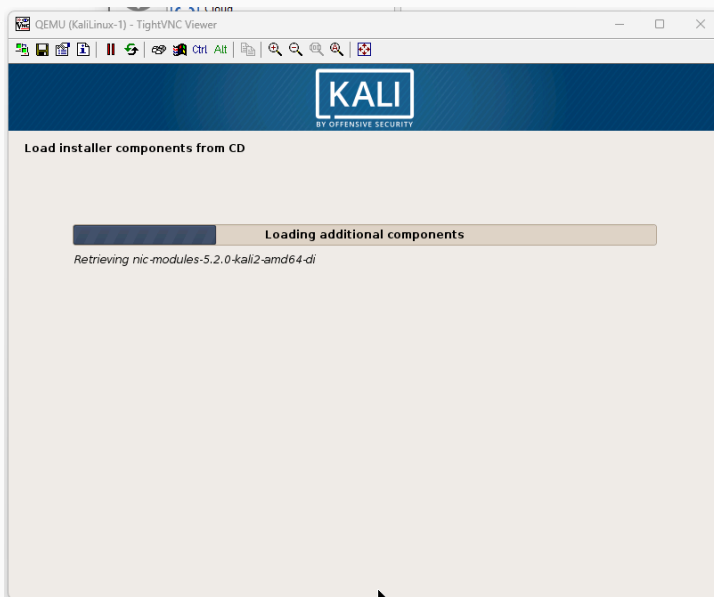
Анализ трафика (Stateless DHCPv6)



“the quieter you become, the more you are able to hear”



Настройка RA + DHCPv6 Stateful (eth2)



Начальные настройки клиента PC3 (до DHCPv6)

QEMU (KaliLinux-1) - TightVNC Viewer

KALI
BY OFFENSIVE SECURITY

Set up users and passwords

You need to set a password for 'root', the system administrative account. A malicious or unqualified user with root access can have disastrous results, so you should take care to choose a root password that is not easy to guess. It should not be a word found in dictionaries, or a word that could be easily associated with you.

A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.

The root user should not have an empty password. If you leave this empty, the root account will be disabled and the system's initial user account will be given the power to become root using the "sudo" command.

Note that you will not be able to see the password as you type it.

Root password:

●●●●

☐ Show Password in Clear

Please enter the same root password again to verify that you have typed it correctly.

Re-enter password to verify:

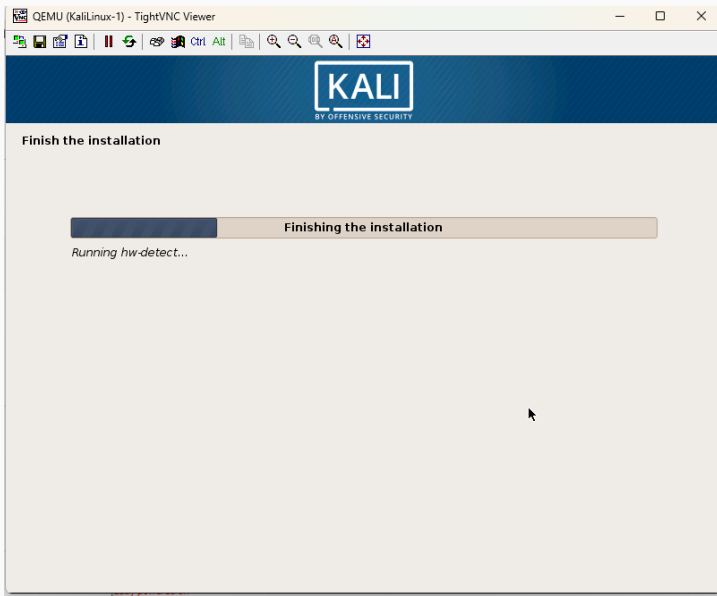
●●●●

☐ Show Password in Clear

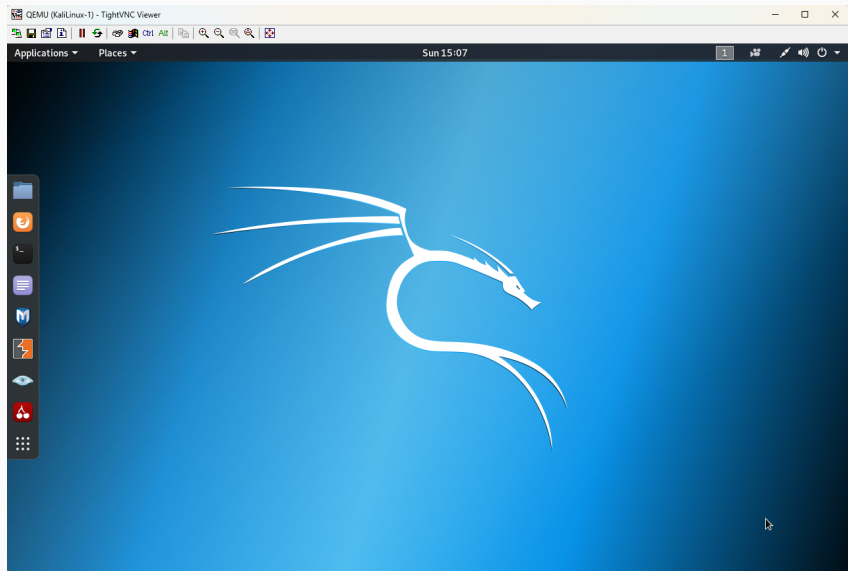
Screenshot

Go Back Continue

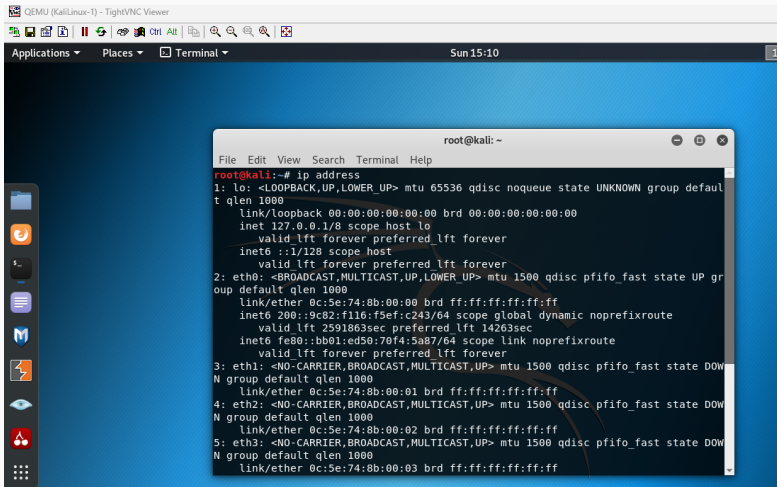
Получение IPv6 по DHCPv6 Stateful (PC3)



Аренды DHCPv6 в Stateful-режиме



Анализ трафика (Stateful DHCPv6)



```
QEMU (KaliLinux-1) - TightVNC Viewer
Applications ▾ Places ▾ Terminal ▾ Sun 15:10 1

root@kali: ~
File Edit View Search Terminal Help
root@kali:~# ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred lft forever
    inet6 ::1/128 scope host
        valid lft forever preferred lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 0c:5e:74:8b:00:00 brd ff:ff:ff:ff:ff:ff
    inet6 200::9c82:f116:f5ef:c243/64 scope global dynamic noprefixroute
        valid lft 2591863sec preferred lft 14263sec
    inet6 fe80::bb01:ed50:70f4:5a87/64 scope link noprefixroute
        valid lft forever preferred lft forever
3: eth1: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast state DOWN group default qlen 1000
    link/ether 0c:5e:74:8b:00:01 brd ff:ff:ff:ff:ff:ff
4: eth2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast state DOWN group default qlen 1000
    link/ether 0c:5e:74:8b:00:02 brd ff:ff:ff:ff:ff:ff
5: eth3: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast state DOWN group default qlen 1000
    link/ether 0c:5e:74:8b:00:03 brd ff:ff:ff:ff:ff:ff
```

Итоги работы

В ходе лабораторной работы: - Развернут стенд в GNS3 с VyOS, коммутаторами и клиентами (VPCS и Kali Linux CLI). - Настроен DHCP для IPv4: - автоматическая выдача адресов, - передача маски/шлюза/DNS/домена, - подтверждена корректность по статистике, арендам, журналу и анализу трафика. - Реализованы два режима DHCPv6: - Stateless: адрес через SLAAC (RA), параметры через DHCPv6, без аренд адресов. - Stateful: адреса выдаются DHCPv6-сервером из диапазона, сервер ведёт учёт аренд. - Анализ захваченного трафика подтвердил корректные сценарии DHCP, DHCPv6, RA и ICMPv6, соответствующие требованиям лабораторной работы.